# Student Performance Management System

## DBMS Project Report

## *Introduction*

The Student Performance Management System is a database-driven application developed using Python and MySQL. The main objective of this project is to manage student-related information efficiently through an interactive graphical user interface (GUI). The system stores and manages student records, attendance, marks, login details, and user activities using a structured relational database.

The project demonstrates the integration of frontend and backend technologies, where Python was used for the application interface and MySQL was used for database management and storage.

## *Technologies Used*

- Python
- MySQL
- MySQL Workbench
- Tkinter (GUI Development)
- GitHub (Project Repository)

## *Features Implemented*

The project includes multiple functionalities for different types of users such as admin, teacher, and students.

### *Main Features*

- User login authentication
- Role-based access system
- Add, update, delete, and search student records
- Marks Management
- Student performance viewing system
- Exporting high performer data into CSV format
- Audit log tracking for user activities
- Database connectivity between Python and MySQL

### *Extra Features Added*

In addition to the basic DBMS requirements, several extra features were implemented to improve the application:

- Separate student login system

- Students can view their own records securely
- Generating report cards for individual students
- Audit logging system for tracking actions
- Improved GUI design for better usability
- Role-based restrictions for users

These additions enhanced both the functionality and security of the application.

## *Database Implementation*

The backend database was created using MySQL. Multiple SQL queries were written to create databases, tables, insert records, and manage relationships between tables.

The project also includes:

- Triggers
- Stored procedures
- Audit log tables
- User authentication tables

The database was exported into an SQL file so that the complete backend can be recreated easily on another system.

## *Use of Artificial Intelligence Tools*

Artificial Intelligence tools were used during the development of this project mainly for GUI improvement, debugging assistance, and correcting certain coding errors. While the database structure, SQL queries, and project logic were developed manually, AI assistance was used to resolve issues related to triggers, frontend integration, and minor code corrections when manual debugging was not sufficient.

AI tools were also used for improving the visual design and usability of the Python GUI application. The use of AI helped in understanding errors more efficiently and completing the project successfully.

## *Conclusion*

This project helped in understanding the practical implementation of DBMS concepts using real-world technologies. It provided hands-on experience in database design, frontend-backend integration, SQL operations, and GUI development. The project also improved problem-solving and debugging skills while demonstrating how database systems can be used to manage and organize student information effectively.

***Link to the GitHub repository***: https://github.com/12rohini12/SPA_db

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